

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Programming Fundamentals and Data Structures	Course Code: CMPE-112L
Assignment Type: Lab Report	Dated: 30th April, 2026
Semester: 1st	Session: 2026
Lab/Project/Assignment #: 12	CLOs to be covered: CLO2
Lab Title: Lambda Functions	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO2	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Mark s	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 12: Lambda Functions

Lab Goals/Objectives:

To understand and use lambda (anonymous) functions in Python as a concise way to define small, single-expression functions.

Equipment Required:

Python 3, VS Code / PyCharm

Theory:

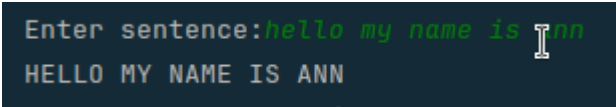
A lambda function is a small anonymous function defined using the lambda keyword. It can take any number of arguments but can only contain a single expression, which is implicitly returned.

Lab Work:**Task 1: Capital and Reverse**

```
string= str(input("Enter sentence:"))
def l_u(answer):
    return answer.upper()
lambda answer: answer.upper()

reverse_string = lambda r: r[::-1]

if str.islower(string):
    text= l_u(string)
    print(text)
elif str.isupper(string):
    print("Reverse:",reverse_string(string))
else:
    print("Please enter a valid sentence")
```

Result:

```
Enter sentence:hello my name is ann
HELLO MY NAME IS ANN
```

Task 2: Lambda - Square

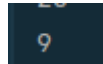
```
a =lambda x: x**2
print(a(5))
```

Result:

Task 3: Lambda - Sum of Two

```
a = lambda x,y: x+y  
print(a(3,6))
```

Result:

A dark-themed terminal window showing the output of the previous code block. The number '9' is displayed in a light blue font.

Task 4: Lambda - Sum of Three

```
a = lambda x,y,z: x+y+z  
print(a(3,6,5))
```

Result:

A dark-themed terminal window showing the output of the previous code block. The number '14' is displayed in a light blue font.

Conclusions:

Lambda functions were used to write concise, single-expression functions, reinforcing an alternative to standard function definitions for simple operations.